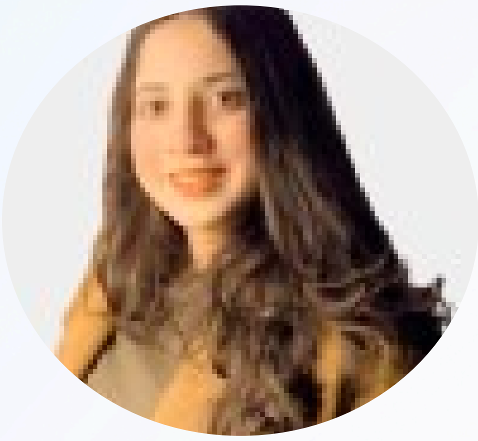

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Dimensional Types in Data Warehousing

1. **Slowly Changing Dimensions (SCDs)**

SCD refers to a way of managing data that changes slowly over time, such as the information about customers or products. For example, if a customer changes their address or their phone number, these are changes that don't happen every day but do occur over time.

There are different ways to handle these changes, and that's where the term SCD comes in. It defines several methods (known as "types") to store and track those changes.

Here are the main types of SCD:

- Type 1 (Overwrite): When something changes, like an address, the old data is simply replaced by the new data. There’s no record of the past value.

- Type 2 (Add a New Record): When something changes, a new record is added to store the new value, and the old value is still kept in the system. This way, you can see the history of changes over time.
- Type 3 (Keep Limited History): Only a limited amount of past data is stored. For example, you may keep the current value and the previous value but not track every change in the future.

2. Conformed Dimension

In a data warehouse, a conformed dimension is a dimension that is used consistently across multiple data marts or different parts of the data warehouse. It means that the same dimension (such as Customer, Product, or Time) is structured and defined in exactly the same way in all areas of the data warehouse, so everyone can use the same data in the same format.

Example:

Let’s say you have two departments in a company: Sales and Marketing. Both departments use customer data, but if they use different formats or categories for customer information, it can cause confusion. A conformed dimension makes sure both departments use the exact same customer details in the same way—so when they look at customer data, they’re both seeing the same thing and can easily compare information.

- How would you handle a situation where you need to update a large number of records in a table but want to minimize locking and performance issues?
- You have a requirement to archive old data from a transaction table to improve performance. How would you approach this, and what considerations would you take into account?
- You need to implement a complex reporting system that requires data from multiple sources and transformations. How would you design the SQL queries or procedures to achieve this?

3. Role-playing dimensions

Role-playing dimensions are dimensions that can be used in multiple ways (or "roles") in different contexts within the same database or reporting system. In other words, the same dimension table is used in different ways depending on the analysis or report being generated.

For example, a Time Dimension could play multiple roles in a data warehouse:

- Order Date: In one report, the Time Dimension might represent when an order was placed.
- Ship Date: In another report, the same Time Dimension might represent when an order was shipped.
- Delivery Date: In yet another report, it could represent when an order was actually delivered.

Even though it’s the same Time Dimension table, it’s being used in different ways in different contexts, depending on the business need.

4. junk dimensions

A junk dimension is a special table in a data warehouse where we store random or miscellaneous information that doesn’t fit neatly into any other table. These are usually small pieces of data (called attributes) that don’t have many different values, so they don’t need their own separate table.

Why do we use junk dimensions?

Sometimes, we have information like:

- Yes/No flags (e.g., "Is the product a gift?"),
- Small categories (e.g., "Payment method" like "Credit Card" or "PayPal"),
- Ratings (e.g., "Customer satisfaction" with values like "Low," "Medium," or "High").

5. Degenerate Dimention

A degenerate dimension is a type of dimension in a data warehouse that doesn't actually have its own separate dimension table. Instead, it's a dimension that exists as a single attribute in the fact table. In other words, a degenerate dimension is just a column of data, often used to represent things like transaction IDs, order numbers, or invoice numbers, that don't require detailed information or a separate table for analysis.

Common Examples of Degenerate Dimensions:

- Order Number: In a sales data warehouse, an Order Number could be stored in the fact table. It's unique for each sale, and you can use it to group or filter data, but you don't need a separate dimension table for just the order number.
- Invoice Number: In a financial data warehouse, an Invoice Number might appear in the fact table, as it's useful for tracking individual transactions but doesn't require its own dimension.

6.Shrunken dimension

A shrunken dimension is a concept in data warehousing that refers to a smaller version of a regular dimension table, containing only a subset of the original dimension's attributes or data. This smaller version is used to improve query performance, especially when you need to analyze data at a higher level of aggregation or a more simplified version of the data.

Example:

If you’re running a sales report and you only care about sales by product category, rather than by individual products, using a shrunken product dimension (with just product categories) instead of the full product dimension (which includes details like size, color, brand) will make your queries faster and simpler.

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